ET AI Hackathon 2026 · Problem 3 · AI Promotion Planner for Indian Retail

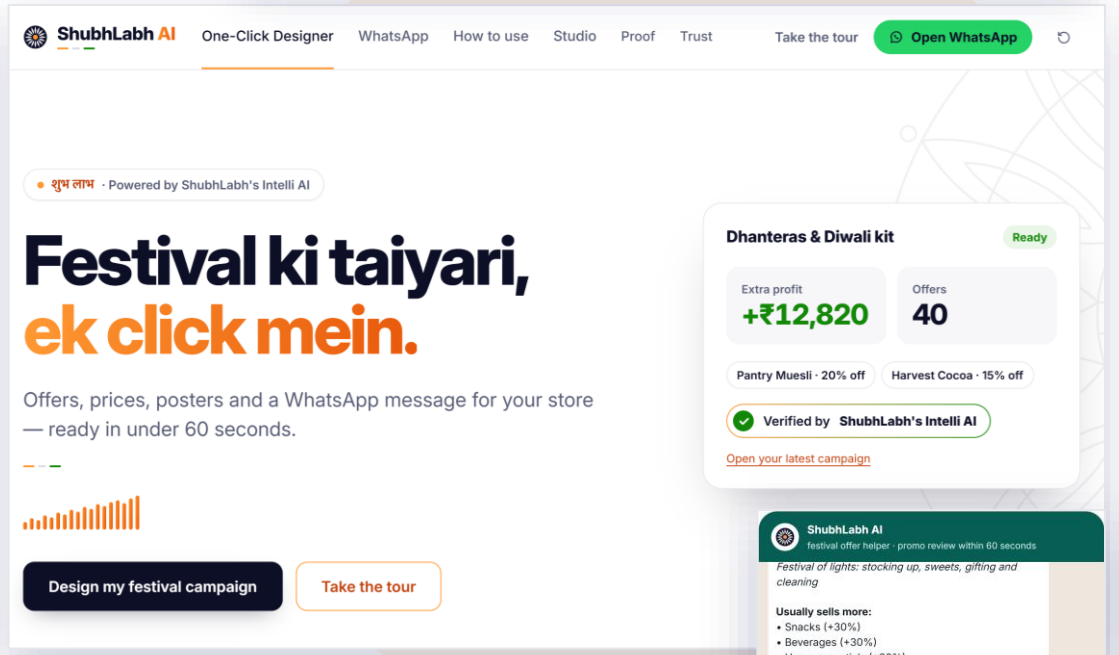
F3 + D3 | all 9 required features + D3 multimodal depth

ShubhLabh AI

India's smartest one-click AI promo planner

Ab sirf laabh hi laabh · अब सिर्फ़ लाभ ही लाभ

One AI promotion engine, three ways in: a **One-Click Campaign Designer** for grocery chains, **Promo Studio** for full control, and **ShubhLabh on WhatsApp** for India's kirana owners.



The app is already live. Use it without any installation:

https://shubhlabh-ai.replit.app

First run may take 4–5 minutes to complete for the One-Click Designer and Promo Studio.

WhatsApp replies instantly: message **+1 415 523 8886** with **'join bright-blanket'**.

Status

Live, working product

F3 + D3 · all 9 features · live on WhatsApp

Built by

Anurag Manik

AI Solution Architect – HP

Team introduction · verified on real data

Built by one person. Verified on real Indian store photos.

AM

Anurag Manik

AI Solution Architect – HP

anuragmanik7@gmail.com

BUILT END TO END, SOLO

- **Backend:** the agent, solver, guard pack and WhatsApp channel
- **React UI:** One-Click Designer, Promo Studio, guided tour
- **Evidence pipeline:** two reads + OCR, Chronos-2 forecasts
- **Proof:** 784 automated tests, red-team runs, real-store checks

REAL STORE CHECK · SAME PIPELINE AS THE PRODUCT

Hand-verified · 10 Oct 2026

Photos & videos from real Indian stores

19 real photos + 2 videos (12 keyframes) from a mall hypermarket, QSR menu boards, apparel and electronics displays, all sent over WhatsApp. Only values at 0.90 confidence or above are auto-accepted; the rest go to a person.

91%
of auto-accepted fields correct — 20 of 22 hand-checkable fields

14 / 14
offer prices correct

80% → 91%
after a consistency check these photos taught us

$0.09
model cost for all 31 inputs · 22 with prices · 45 extra items read



Super Sarvottam Rice Bran Oil 13 kg: MRP ₹2,929 → ₹2,199



KFC Krisper Meal ₹237 → ₹99



Caught by review: "33% off" read as ₹333, sent to a person

Labelled test set, live re-run (61 files): 100% of auto-accepted fields correct on all 6 input types — shelf-tag photos, festival flyers, quick-commerce screenshots, WhatsApp scheme images, text-PDF and scanned price lists (slide 9).

Problem statement

Festivals decide the year — and most promotions lose money

₹6.05 lakh cr

Diwali 2025 trade, +25% on 2024 · 85% of it through traditional retail

— CAIT, Oct 2025

~91%

of India's US$658 bn grocery market is sold through 12–13 million kiranas

— Redseer, 2026 · Invest India

59%

of trade promotions lose money (72% in the US)

— McKinsey, 2019 (citing Nielsen)

22%

of companies can measure promotion spend per event · 81% run compliance by hand

— Strategy& (PwC) · POI, 2026

FIVE PAIN POINTS

P1 Evidence is scattered and messy

Schemes, competitor prices and HQ rules arrive as WhatsApp forwards, shelf photos, flyers and PDFs. Nobody reads them all, and one hidden instruction can mislead an AI.

500 M+ WhatsApp users in India — Meta

P2 Planning is slow and unproven

Weeks of spreadsheets per scheme, no forecast, and no proof an offer will pay before it goes live.

22% can measure spend per event — Strategy&

P3 Quick commerce resets prices daily

Ten-minute apps run 20–25% off MRP city by city; chains and kiranas must answer in hours, on margins of 10–12%.

Elara Capital, 2025 · KPMG, 2026

P4 Compliance risk is now real

Fake "% off", drip pricing and MRP breaches carry penalties; from 1 Jan 2027 every online price cut must show the 30-day low.

CCPA fines on 9 platforms, Aug 2026 · E-Commerce (Amdt) Rules, 2026

P5 The kirana has no planner at all

12–13 million shops sell ~91% of India's grocery with no analyst, no tool and no app they would install.

Redseer, 2026 · Invest India

WHAT THAT MEANS FOR ONE BRAND

A ₹500-crore consumer brand puts ~₹100 crore a year into promotions. If 59% of those events lose money, crores ride on gut feel every festival.

Decided blind — in spreadsheets, days too late

Proposed solution · Powered by agentic AI

One-Click Designer · speed, for chains

Promo Studio · full control, for category & promo managers

WhatsApp · for kiranas

One AI engine, three ways in — every pain point answered

P1	Messy evidence	**Reads anything** — D3 Shelf-tag photos, flyers, app screenshots, WhatsApp schemes, PDFs, CSVs. Two independent reads + OCR; doubtful fields go to a person; hidden instructions quarantined.
P2	Slow, unproven planning	**One click in under 2 minutes, or full control in Promo Studio** — Hero USP 1 One-Click: festival template → Chronos-2 forecast → agent designs offers → Intelli AI verifies each one → kit. Promo Studio: your own brief, every offer's checks, sign-off tiers, live what-ifs.
P3	Daily price resets	**Forecasts and reacts** — Live ops Demand per product × city with P10–P90; competitor prices read from photos; live events (competitor cut, stock drop, budget cut) replan mid-festival with a diff.
P4	Compliance risk	**Verifies every rupee** — Guard pack Never above MRP · honest % off against the 30-day low · same price for everyone · dark-pattern lint · checked by ShubhLabh's Intelli AI, with a confidence score.
P5	No planner for kiranas	**ShubhLabh on WhatsApp** — Hero USP 2 Message +1 415 523 8886 or scan the QR: festival tips, an offer check with margin and break-even, a competitor price-tag photo read. English or Hinglish. No app, no login — you don't need the website.

AI designs. ShubhLabh's Intelli AI verifies every rupee. A person approves. Built for WhatsApp — where India's kiranas already are.

Architecture · Powered by agentic AI

Six layers: evidence in, verified offers out

1	**Inputs** anything the team has	Brief in English or Hinglish	Shelf-tag photos · flyers · app screenshots · WhatsApp images	HQ policy & price-list PDFs	Sales · stock · competitor CSV / Excel	WhatsApp messages from kiranas
2	**Reading & verification** two reads must agree	Vision read — gpt-4.1-mini, strict JSON, no tools	RapidOCR local read + parser	Agree + locate → confidence · ≥ 0.90 auto-accept, else a person	Injection quarantine · PII redaction	Brief compiler → typed constraints
3	**Forecasting & optimisation** deterministic code	Chronos-2 demand forecast per product × city · P10–P90	Price response calibrated on 4.85M real records	Candidates: markdown · coupon · bundle	OR-Tools CP-SAT solver + recovery ladder	Monte Carlo risk check · 200 scenarios
4	**Intelli AI verification** every offer, every rupee	Never above MRP	Honest % off vs 30-day low	Same price for everyone	Margin floor · stock · budget	Dark-pattern lint	Number check	Verification record per offer
5	**Approval** a person in the lead	Approval tiers · senior sign-off above ₹15,000 worst case or 20% depth	Ask-human mid-run	Live-event triage → replan with a diff	Full audit trail · versioned plans
6	**Channels** where the plan goes	One-Click: campaign kit · calendar · channels · English + Hindi copy · banners	Promo Studio: plan · every check · sign-off · live replans	WhatsApp: replies via an n8n automation	Approved price feed (JSON) · exports

AGENT ORCHESTRATOR · LAYERS 2–5

AGENT ORCHESTRATOR · gpt-4.1-mini with function calling, across layers 2–5

Chooses the next tool and logs why · recovers (solver time-out → prune → split → fallback; model down → rules policy) · asks the manager when a brief contradicts itself · can never approve its own plan

STACK

FastAPI · SQLAlchemy · PostgreSQL (SQLite on a laptop) · React · n8n for WhatsApp · OpenAI with a hard spend cap · works rules-only without a key

AI models & technologies used

Each model does one job — and is measured at it

Model / technology	What it does in ShubhLabh	Why this one	Measured
OpenAI gpt-4.1-mini	Vision read of evidence (strict JSON, no tools) and targeted re-reads of doubtful fields; compiles briefs into constraints; agent tool choice; live-event triage; copy and explanations	Fast and low-cost with function calling; every call logged; hard daily spend cap	≈ $0.011 model cost per AI plan · 100% of auto-accepted fields correct on the 61-file labelled set
Chronos-2 time-series foundation model (local CPU)	Demand forecast per product × city with P10–P90 over a 14-day horizon; discount and promo activity as known covariates	Zero-shot: no training on the store; runs on a CPU, no GPU	WAPE 36.4% vs 40.0% seasonal naive · 600 real series
RapidOCR ONNX, local	Second, independent read of every image and scan; locates each value on the page (boxes)	Free and offline; the model read and the OCR read must agree before auto-accept	Real store photos: 20 of 22 auto-accepted fields correct · offer prices 14 / 14
OR-Tools CP-SAT solver	Chooses offers under budget, stock, margin floor, cannibalisation and city rules; recovery ladder hint → prune → split → fallback	Exact optimisation, not a heuristic; every result re-validated independently of the model	120 SKUs × 3 cities, 848 candidates: P95 1.34 s
Monte Carlo risk check	Runs plan, flat 10%, conservative and no-promotion on the same 200 scenarios; P10 / P50 / P90 and downside probability	Independent of the solver; selection and reporting on separate scenario seeds	Plan 2.0× a flat 10% off · at least ₹10,201 nine times in ten
Price-response model	Elasticity and promo-activity lift behind every candidate price and every forecast scenario	Calibrated on real retail demand, not assumptions	4.85M records · 898 stores · 18 cities · 62% smaller loss than flat 10% in a back-test
ShubhLabh's Intelli AI verification layer	Our own in-house AI agent. In the background it re-checks every calculation on every offer: MRP, honest % off vs the 30-day low, margin, break-even and budget, plus copy lint and injection quarantine. It then gives a confidence score: a 90% chance of at least its stated extra profit, from the 200-scenario risk check.	Every published offer is verified by ShubhLabh's Intelli AI, with a confidence score	190 / 190 red-team cases caught across 7 guards · 0 false alarms
Agentic orchestration	Tool calls with strict schemas; picks the next tool and logs why; recovers from solver time-outs and model outages; asks a person when unsure	Powered by agentic AI, with a person approving every plan	Rules-only policy with no key · 784 automated tests

STACK FastAPI · SQLAlchemy · PostgreSQL (SQLite on a laptop) · React + Vite · n8n automation for WhatsApp · NumPy · scikit-learn · pytest (784 tests) · Python

Product Demo

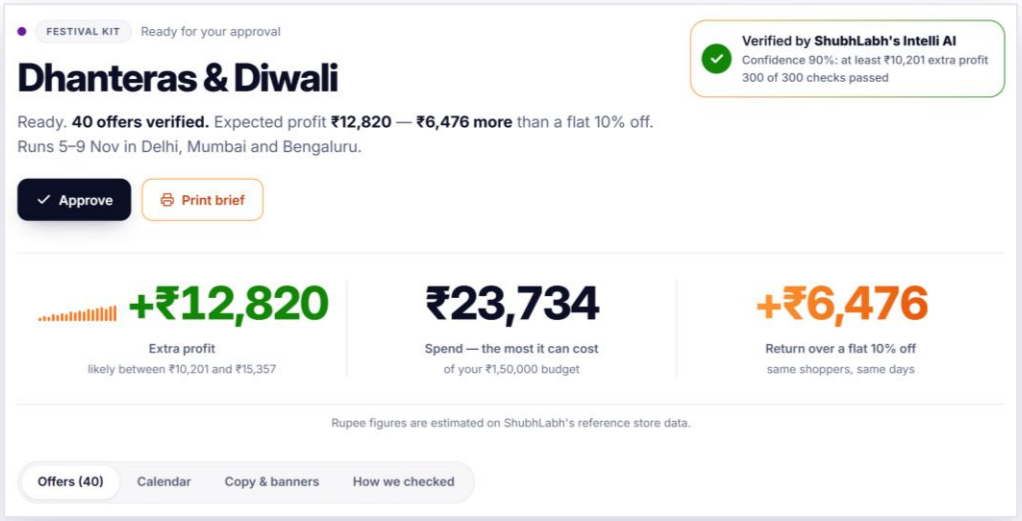
Live: shubhlabh-ai.replit.app

Under 2 minutes

One click. Full control. One message.

ONE-CLICK DESIGNER · SPEED, FOR CHAINS

Pick a festival → Drop in anything → Press Design → Verified kit · <2 min

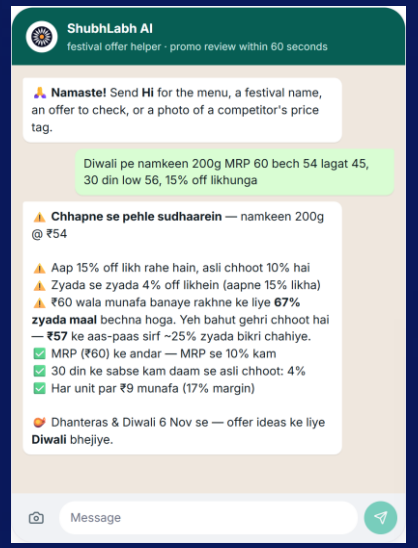


Dhanteras & Diwali · Delhi · Mumbai · Bengaluru · +₹12,820 extra profit · 2.0× a flat 10% off · 40 offers verified by ShubhLabh's Intelli AI · 300 of 300 checks passed · estimated on ShubhLabh's reference store data (rules-only run)

Zero training: a guided tour on the first visit and three How-to guides with ~70-second Hinglish videos. The full walkthrough video is in the submission.

Full control in Promo Studio → next slide: brief, evidence, agent run, plan & approve, live ops, ask

SHUBHLABH ON WHATSAPP · LIVE



Offer check: MRP · honest % off · margin · break-even

1. Message +1 415 523 8886 — or scan the QR
2. Send "join bright-blanket" — typed in for you by the Open WhatsApp button
3. Then send an offer, a festival or a price-tag photo

Scan to try it now
You don't need the website.

English or Hinglish · No app, no login · Phone numbers never stored · Hidden instructions quarantined · Only ever replies, never messages first

Promo Studio · for category & promo managers

Promo Studio: full control, every step verified

1 Brief

YOUR WORDS → EXACT CONSTRAINTS

Budget | Discount cap | Margin floor | Cities | Categories | Mechanisms | Days

Read back to you before anything runs

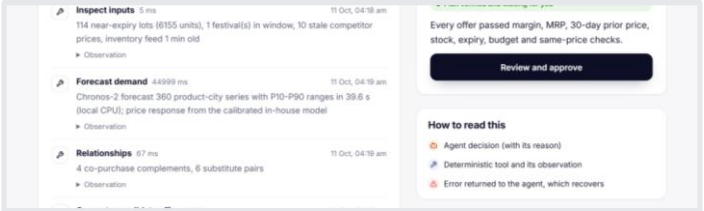
Write it in plain English or Hinglish. It is compiled into exact constraints and read back to you before anything runs.

2 Evidence



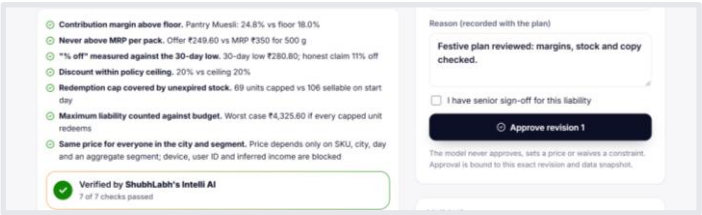
Shelf photos, flyers, app screenshots, PDFs, CSVs. Two reads must agree, with a box on every value read; doubtful fields go to a person; hidden instructions quarantined.

3 Agent run



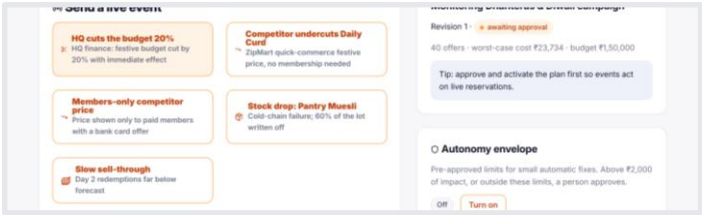
Watch the agent work step by step, with every tool call, decision and recovery logged.

4 Plan & approve



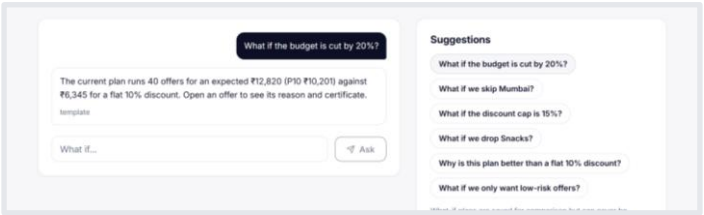
Offers by city and segment. Open each offer's checks, verified by ShubhLabh's Intelli AI. Senior sign-off tiers; nothing goes live without approval.

5 Live ops



Competitor undercut, stock drop or HQ budget cut → replan with a diff, then approve.

6 Ask



What-ifs in plain words: cut the budget, cap the discount, drop a category, skip a city, with before and after.

What we do differently

Built for Indian retail, not adapted to it

Solution	Indian festival templates	Reads WhatsApp & shelf photos	Every offer verified (MRP · 30-day low)	One-click campaign	Hindi / Hinglish	Works for a kirana on WhatsApp
Enterprise promo suites Global, ₹-crore licences	⊖	⊗	⊖	⊖	⊗	⊗
BI dashboards & Excel Today's default	⊗	⊗	⊗	⊗	⊗	⊖
Marketplace & q-commerce seller tools Platform-locked	⊖	⊗	⊗	⊖	⊖	⊗
Generic AI chatbots Unverified answers	⊖	⊖	⊗	⊗	⊖	⊖
ShubhLabh AI	✓	✓	✓	✓	✓	✓

WHY IT IS HARD TO COPY

- Two independent reads must agree before any price is used
- Every offer verified by Intelli AI — not a dashboard, not a chatbot answer
- Price response calibrated on 4.85M real retail records

Comparison by category from publicly available product information, October 2026.

Requirement coverage

All 9 required features — plus D3

F3 + D3

all 9 features · multimodal depth

Product selection
Screens 120 SKUs to the best 20, with reasons

Promotion mechanism
Markdown, coupon or bundle per product and city

Discount & strategy
Depth, timing, audience, city — each verified

Cannibalisation
Substitute effects charged to every offer

Product relationships
AI suggests pairs; data confirms them

Inventory constraints
Stock, batches, expiry; near-expiry first

Geographic customisation
A different plan per city

Competitor awareness
Prices read from photos, compared per 100 g

Test before launch
Plan vs flat 10% vs none, 200 scenarios

D3 · Multimodal depth

LIVE RE-RUN · AUTO-ACCEPTED FIELDS CORRECT

Shelf-tag photos	100%
Festival flyers	100%
Quick-commerce screenshots	100%
WhatsApp scheme images	100%
Text-PDF price lists	100%
Scanned price lists	100%

61 labelled files · coverage 100% · two reads + OCR · boxes on values read from images · doubtful fields go to a person

85%+ raw field accuracy
with no model key (rules-only; then every field goes to a person)

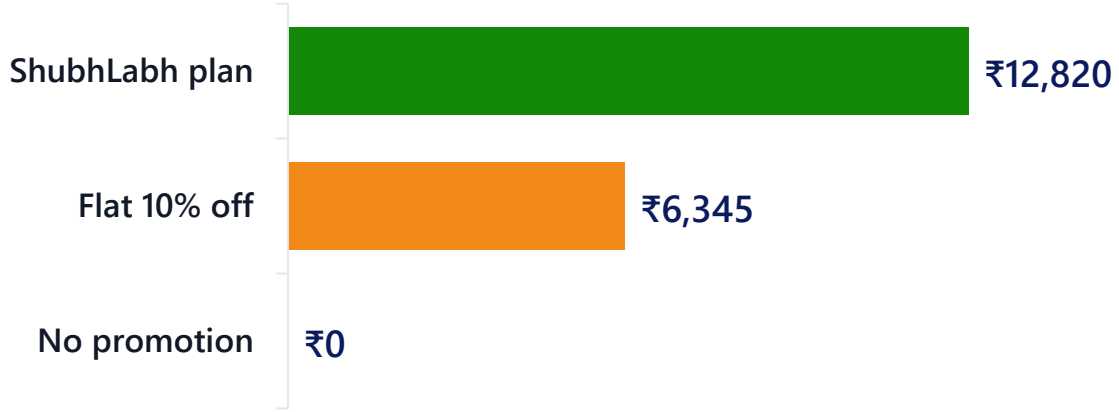
Every feature runs end to end in the live product · every published offer verified by ShubhLabh's Intelli AI · plus WhatsApp for kiranas

Real store photos (slide 2): 20 of 22 hand-checkable auto-accepted fields correct (91%) · offer prices 14 of 14 · a hidden instruction in a competitor PDF quarantined before any model read it.

Business impact

2.0× the extra profit of a flat discount — and not one offer above MRP

Diwali campaign · extra profit on the same 200 forecast scenarios



At least ₹10,201 nine times in ten · 1,410 near-expiry units cleared · costs at most ₹23,734 of a ₹1,50,000 budget. Estimated on ShubhLabh's reference store data (rules-only run); price response calibrated on 4.85M real retail records.

<2 min

typically, from one click to a verified campaign

Measured in the product · 44–92 s in four of five runs

2.0×

the extra profit of a flat 10% discount

Estimated on ShubhLabh's reference store data

1,410

near-expiry units cleared in one campaign

Estimated · less waste, fewer write-offs

0

offers above MRP or with a fake "% off"

Verified by ShubhLabh's Intelli AI · 190/190 red-team cases caught

DOOR 1 · CHAINS

Shadow pilot first (nothing goes live) → one festival end to end in 2–3 cities → always-on weekly plans. **Revenue:** SaaS per store per month + campaign credits.

DOOR 2 · KIRANAS

Free offer checks and festival tips on WhatsApp, sponsored by FMCG brands and distributors who already run schemes to kiranas; premium plans. **≈ $0.011 of AI cost per plan keeps it free.**

Scalability · Future roadmap

Light, cheap to scale — next: every Indian store

SCALABILITY · MEASURED ON THE DEPLOY AUDIT (11 OCT 2026)

Multi-tenant PostgreSQL
A workspace per chain or shop; the full test suite passes on PostgreSQL 17

Light, no GPU
~1 GB RAM with Chronos-2 on CPU; one VM runs forecasting, OCR and the solver

Solver headroom
120 SKUs × 3 cities, 848 candidates, solved in P95 1.34 s

Cheap AI, capped
≈ $0.011 of model cost per plan, under hard daily and total spend caps

Ready for a crowd
Rate limits per client and per WhatsApp sender; CSRF and CSP on every route

Fast and tested
Kit in ~40 s with Chronos-2, 6.5 s without · 784 automated tests

LIVE TODAY

- **One-Click Campaign Designer · 12 festival templates**
- **Promo Studio · full control from brief to sign-off, live what-ifs**
- **ShubhLabh on WhatsApp · +1 415 523 8886**
- **English + Hindi copy · Hinglish chat**
- **Live-event replanning with a diff**
- **PostgreSQL · approval tiers · full audit trail**

NEXT 3 MONTHS

- Shadow pilots with 2–3 grocery chains — nothing goes live
- Pilots with FMCG brands and distributors who sponsor kirana plans
- A chain's or store's own WhatsApp Business number
- Recalibrate price response on Indian pilot outcomes

3–6 MONTHS

- POS / billing / ERP connectors — sales, stock and MRP sync, no CSVs
- More Indian languages: Tamil, Telugu, Bengali, Marathi, Kannada, Gujarati
- Regional festivals — Onam, Durga Puja, Eid, Ganesh Chaturthi, Raksha Bandhan — and more cities

6–12 MONTHS

- Scheduled always-on weekly plans, auto-briefed from last week's sales
- Outcome loop: redemptions and sales feed the next plan
- Background job queue so the engine runs across several servers
- Voice notes on WhatsApp

Ab sirf laabh hi laabh.

अब सिर्फ़ लाभ ही लाभ

Anurag Manik · AI Solution Architect – HP

anuragmanik7@gmail.com

Live: shubhlabh-ai.replit.app · F3 + D3 · repository in the submission

Scan to try it on WhatsApp

"join bright-blanket" typed in for you